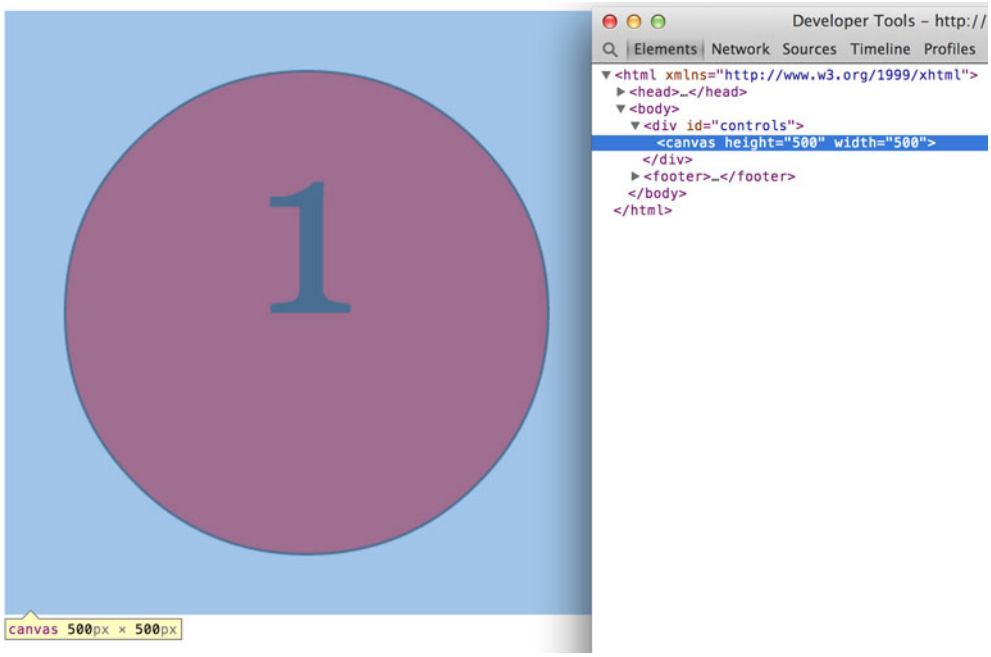


Figure 8.7 Any graphics created in canvas are stored as a bitmap or raster image. Unlike in SVG, the individual shapes are no longer accessible or modifiable after being drawn.

element, but on a `<canvas>` element that needs to be created in the DOM. Third, canvas has a syntax distinct from SVG.

But there's one more major difference between the graphics created using canvas and the graphics created using SVG. You can see it if you inspect that circle, as shown in figure 8.7. Anything drawn in canvas is drawn to a bitmap, so you don't have an individual text or circle element that you could assign an event listener to, or whose appearance or text content you can later modify. It's also not vector-based, so if you try to zoom the image, you'll see the pixilation you're familiar with from zooming photos and other raster imagery. Because HTML5 canvas doesn't create separate DOM elements, it benefits from higher performance when dealing with large amounts of those graphical elements. But you lose the flexibility of SVG.

8.3.2 Drawing and storing many images

We want images because our plan is to build an image gallery, but the canvas element in the DOM doesn't act like the kind of image that you're accustomed to dealing with in web development. You can't right-click and save it, or open it in a new window in its current form. But the `<canvas>` element includes a `.toDataURL()` function that provides a string designed to be the `src` attribute of an `<img>` element. You can see in the following listing the results of `.toDataURL()` when applied to one of your drawn circles. This is only the first three lines—the actual value would go on like this for nine pages.